# Week 1 Task Update Report

**Progress Completed:**

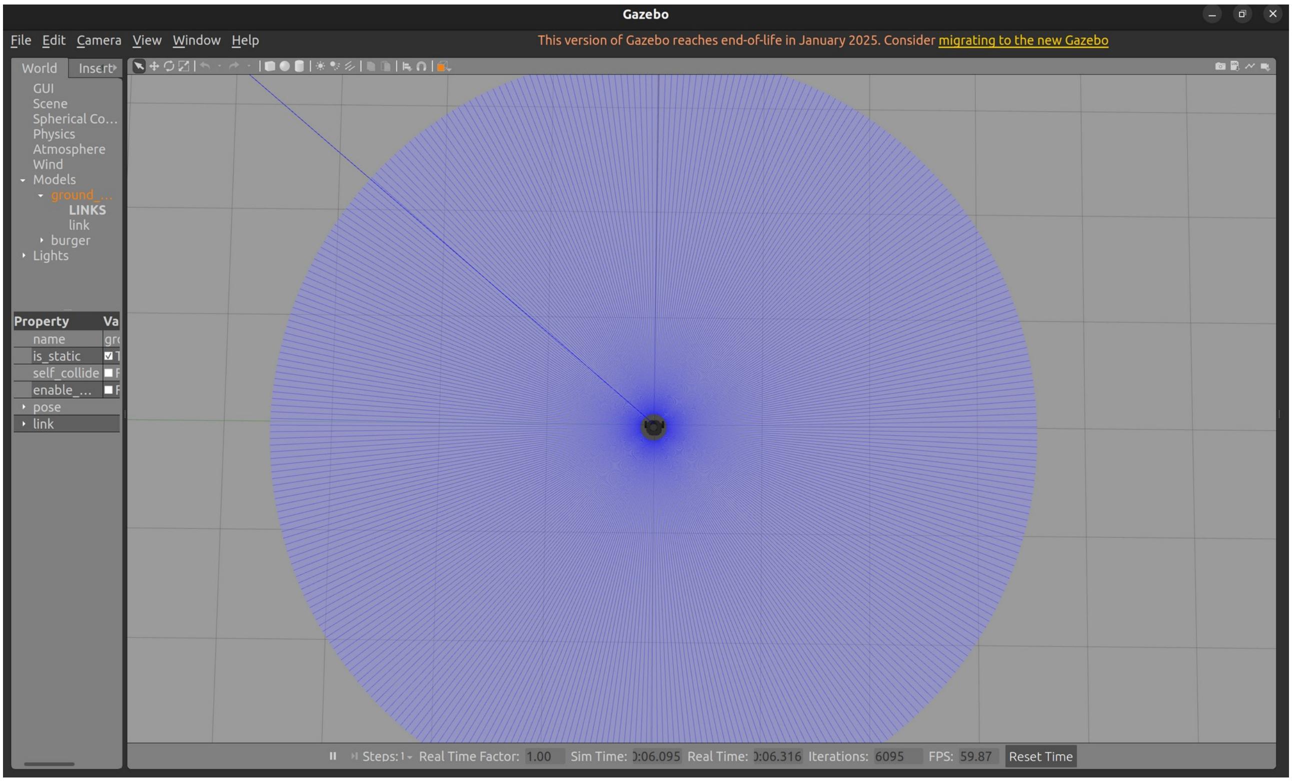
- Successfully provisioned an Ubuntu 22.04 environment and installed ROS 2 Humble Hawksbill alongside the Gazebo physics simulator.
- Configured the TurtleBot3 simulation packages (turtlebot3_gazebo) and successfully launched the empty_world environment.
- Installed the VirtualIMU APK on an Android device and verified local Wi-Fi network connectivity.
- Wrote a foundational Python script utilizing the socket library to successfully bind to port 1212, receive UDP datagrams from the mobile application, and parse the delimited UTF-8 string into discrete Accelerometer, Gyroscope, and Magnetometer arrays.

**Challenges Faced:**

- **Network Firewall Restrictions:** Initially, the UDP packets were being dropped. I resolved this by utilizing ufw to explicitly open port 1212 for incoming UDP traffic and verifying the stream using netcat before connecting the Python script.
- **Coordinate System Mismatch:** Discovered that raw accelerometer data mapping differs between standard aeronautical models and Android hardware (where the Y-axis is vertical).

**Plans for the Next Week (Week 2):**

- Migrate the standalone Python socket script into a fully functional ROS 2 rclpy Node.
- Implement the mathematical kinematic transformations to convert raw accelerometer data (m/s^2) into Pitch and Roll angles using math.atan2.
- Map these calculated angles to a geometry_msgs/Twist message and publish them to the /cmd_vel topic to achieve real-time differential drive control in Gazebo.

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rasal@rasal:~/Documents/crob_ws$ nc -u -l 1212
0.270,0.337,9.693|-0.007,-0.013,-0.000|-39.737,-12.863,-11.2000.258,0.349,9.729|-0.006,-0.01
3,0.001|-39.737,-12.863,-11.2000.270,0.349,9.700|-0.004,-0.009,0.001|-39.737,-12.863,-11.200
0.241,0.363,9.690|-0.000,-0.001,0.001|-39.737,-12.863,-11.2000.256,0.344,9.717|-0.005,-0.007
,-0.001|-39.725,-12.913,-11.1750.224,0.359,9.688|0.001,0.007,-0.000|-39.737,-12.863,-11.2000
.260,0.361,9.681|-0.004,-0.006,0.002|-39.725,-12.913,-11.1750.224,0.359,9.702|0.001,0.005,0.
001|-39.725,-12.913,-11.1750.246,0.340,9.719|0.001,0.002,-0.001|-39.725,-12.913,-11.1750.256
,0.347,9.678|-0.002,-0.001,0.001|-39.725,-12.913,-11.1750.229,0.349,9.695|0.001,0.004,0.002|
-39.725,-12.913,-11.1750.217,0.347,9.698|-0.000,0.003,0.001|-39.725,-12.950,-11.1500.220,0.3
83,9.698|0.001,0.004,0.001|-39.737,-12.863,-11.2000.239,0.356,9.700|-0.004,-0.005,-0.001|-39
.725,-12.913,-11.1750.222,0.387,9.702|0.002,0.009,-0.000|-39.737,-12.863,-11.2000.217,0.356,
9.662|-0.000,-0.002,0.001|-39.725,-12.913,-11.1750.222,0.363,9.693|0.002,0.010,-0.000|-39.73
7,-12.863,-11.2000.215,0.356,9.683|-0.000,-0.002,-0.000|-39.725,-12.950,-11.1500.248,0.342,9
.717|-0.002,-0.009,-0.001|-39.725,-12.950,-11.1500.246,0.347,9.669|-0.004,-0.003,-0.000|-39.
725,-12.950,-11.1500.241,0.351,9.702|-0.002,-0.001,-0.000|-39.725,-12.950,-11.1500.248,0.340
,9.700|-0.004,-0.005,0.001|-39.725,-12.950,-11.1500.244,0.354,9.678|-0.002,-0.006,-0.000|-39
.725,-12.950,-11.1500.251,0.349,9.724|-0.004,-0.009,-0.001|-39.725,-12.950,-11.1500.229,0.34
4,9.722|-0.000,0.001,-0.001|-39.725,-12.975,-11.1630.227,0.342,9.700|-0.002,-0.001,-0.000|-3
9.725,-12.975,-11.1630.241,0.351,9.683|-0.002,-0.005,-0.001|-39.725,-12.975,-11.1630.210,0.3
61,9.714|-0.002,-0.003,-0.000|-39.725,-12.975,-11.1630.215,0.356,9.693|-0.002,-0.001,-0.001|
-39.725,-12.975,-11.1630.232,0.351,9.717|-0.004,-0.004,0.001|-39.725,-12.975,-11.1630.224,0.
356,9.722|-0.002,-0.002,0.001|-39.725,-12.975,-11.1630.227,0.361,9.686|-0.002,-0.002,0.001|
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